Publications

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Research Areas:
Bioanalytical Chemistry • Glycoproteomics • Protein Enzymology • Structural Biology • Crystallography • Mass Spectrometry • Biopharmaceutical Innovation • Sustainable Analytical Method Development

Peer-Reviewed Journal Publications

2026

- **Ajayi, O. O.**, & Aduloju, T. A. (2026).
  Protocols for Assessing Storage Stability and Solubility of Released N-Linked Glycans Prior to Mass Spectrometry Analysis.
  Current Protocols, 6, e70509. https://doi.org/10.1002/cpz1.70509

2025

- **Ajayi, O. O.**, & Ajayi, V. O. (2025).
  A Systematic Optimization Analysis of N-Linked Glycans on Porous Graphitic Carbon (PGC) at Different Column Temperatures.
  Journal of Progress in Engineering and Physical Science, 4(6), 22-29.
  https://doi.org/10.63593/JPEPS.2025.12.04
- Aduloju, T. A., Ezekwesili-Ofili, J. O., **Ajayi, O. O.**, & Oladejo, A. A. (2025).
  Potential of Pumpkin (*Cucurbita maxima*) Fruit Used as Functional Food for the Management of Diabetes and Dyslipidemia.
  European Journal of Nutrition & Food Safety.
- Aduloju, T. A., Ezekwesili-Ofili, J. O., Oladejo, A. A., & **Ajayi, O. O.** (2025).
  Modulation of Glycemic and Lipid Metabolic Pathways by *Cucurbita maxima* Fruit Parts in Streptozotocin-Induced Diabetic Rats.
  Journal of Complementary and Alternative Medical Research.
- Ajayi, V. O., Ameh, R. F., & **Ajayi, O. O.** (2025).
  Application of Predict Explain Observe Explain Worksheet in Reconstructing Students' Misconceptions in the Context of Alkanes and Alkanols.
  The Journal of Educational Paradigms.
- Ajayi, V. O., Ameh, R. F., Chinasa, G. P., & **Ajayi, O. O.** (2025).
  Assessing Students' Misconceptions and HOTS Levels in the Context of Chemical Reaction Using Paper-Based Five-Tier Test Diagnostic Instrument.
  Caribbean Journal of Education.

2024

- **Ajayi, O. O.** (2024).
  A Review of Mass Spectrometry as an Important Analytical Tool for Structural Elucidation of Biological Products.
  Journal of Progress in Engineering and Physical Science, 3, 23-31.

Published Abstracts

- **Ajayi, O. O.**, & Phillips, R. S. (2025).
  Investigating Cofactor Incorporation in AprL Aminotransferase: Redox, Chemical, and Cofactor Form Dependencies.
  Free Radical Biology and Medicine. Annual meeting abstract, Society for Redox Biology and Medicine.

Preprints

- Ajayi, V. O., Ameh, R. F., Sor, E. N., & **Ajayi, O. O.** (2025).
  Science Educators' Challenges of Fostering Students' Essential Skills for Global Citizenship in Public Universities in Kogi State, Nigeria.
  SSRN Electronic Journal.

Book Chapters in Editorial Production

First author of all chapters. Invited book chapters currently under editorial review and/or print production with Springer Nature (Switzerland), CRC Press/Taylor & Francis (USA), and Routledge/CABI (United Kingdom).

- **Ajayi, O. O.,** Ajayi, V. O., Aisedion, M., & Penda, B. M.
  Chemical Industry: Biocatalysis and Enzyme Production, Synthesis of Fine Chemicals, and Green Chemistry and Sustainable Processes.
  In *Prospects of Engineered Bacteria, Volume 2: Engineered Bacteria: Applications in Industry, Chapter 6.*
  Routledge/CAB International, United Kingdom.
  Status: Submitted for publication; currently in editorial production.
- **Ajayi, O. O.,** Ajayi, V. O., & Aisedion, M.
  Processed Development of Microbial Cell Lines in Biopharmaceuticals: Innovations, Challenges, and Future Directions.

In *Advances and Applications in Biotechnology, Volume 5: Biotechnology of Cell Lines in Biopharma, Chapter 2.*
CRC Press/Taylor & Francis, USA.
Status: Submitted for publication; currently in editorial production.

- **Ajayi, O. O.,** Ajayi, V. O., & Aisedion, M.
  An Alternative Strategy for the Use of Antimicrobials in Animals.
  In *Prospects of Fungal Biotechnologies for Livestock, Volume 4: Fungal Proteins and Enzyme Processes in Animal Husbandry, Chapter 9.*
  Springer Nature, Switzerland.
  Status: Submitted for publication; currently under editorial review.
- **Ajayi, O. O.,** Ajayi, V. O., Salami, B. A., & Ayodeji, S. T.
  Biochemical Reconstitution of AprL Aminotransferase: A Synthetic Biology Framework for PLP-Dependent Enzyme Engineering in Bacterial Systems.
  In *Prospects of Engineered Bacteria, Volume 1: Engineered Bacteria: Applications in Agriculture, Chapter 11.*
  Routledge/CAB International, United Kingdom.
  Status: Submitted for publication; currently under editorial review.
- **Ajayi, O. O.,** Ajayi, V. O., & Aisedion, M.
  Fungal Biotechnologies for Enhanced Protein Quality in Livestock Feed: Advances and Prospects.
  In *Prospects of Fungal Biotechnologies for Livestock, Volume 3: Myco-Engineered Food for Livestock, Chapter 11.*
  Springer Nature, Switzerland.
  Status: Submitted for publication; currently under editorial review.
- **Ajayi, O. O.,** Ajayi, V. O., & Aisedion, M.
  Enzyme-Responsive Delivery Platform: Focusing on Tumor-Specific Enzymes.
  In *Chemical Modulators of the Tumor Microenvironment: A Pharmaceutical Chemistry Perspective, Section 2: Pharmaceutical Chemistry Tools for Tumor Microenvironment Modulation, Topic 6.*
  Status: Submitted for publication; currently under editorial review.

Research Software

Sole author of all packages. Open-source Python software released under the MIT license and archived on Zenodo with a DOI. Manuscripts based on Solubility Noise Benchmark and Raman Bioprocess Chemometrics are submitted (see Submitted Manuscripts); a software paper for each of the first six packages is in preparation.

- **Ajayi, O. O.** (2026).
  Laboratory Automation Digital Twin Framework: An Open-Source Platform for

Modeling Material Transport, Device Coordination, and Sample Traceability in Automated Laboratory Workflows.
Zenodo. Version 2.0.0 [Computer software]. https://doi.org/10.5281/zenodo.23556262

- **Ajayi, O. O.** (2026).
Automation Readiness Assessment Framework: An Open-Source Methodology for Evaluating Laboratory Workflow Automation Readiness.
Zenodo. Version 2.0.0 [Computer software]. https://doi.org/10.5281/zenodo.23556257
- **Ajayi, O. O.** (2026).
SDL Framework: A Reproducible Python Framework for Closed-Loop Experimental Optimization Using Self-Driving Laboratory Principles.
Zenodo. Version 1.1.1 [Computer software]. https://doi.org/10.5281/zenodo.23550223
- **Ajayi, O. O.** (2026).
LC-MS Data Processor: An Open-Source Python Workflow for Chromatogram Processing, Peak Integration, Calibration, and Quality-Control Reporting.
Zenodo. Version 1.1.1 [Computer software]. https://doi.org/10.5281/zenodo.23550222
- **Ajayi, O. O.** (2026).
Enzymatic Kinetics Analyzer: An Open-Source Workflow for Reproducible Enzyme Kinetics and Dose-Response Analysis.
Zenodo. Version 2.0.1 [Computer software]. https://doi.org/10.5281/zenodo.23550220
- **Ajayi, O. O.** (2026).
Drugability Toolkit: An Open-Source RDKit-Based Workflow for Interpretable Small-Molecule Developability Assessment.
Zenodo. Version 1.1.1 [Computer software]. https://doi.org/10.5281/zenodo.23550217
- **Ajayi, O. O.** (2026).
Solubility Noise Benchmark: Molecular Property Models Under Label Noise and Scaffold Shift.
Zenodo. Version 0.1.1 [Computer software]. https://doi.org/10.5281/zenodo.23308185
- **Ajayi, O. O.** (2026).
Raman Bioprocess Chemometrics: PLS Calibration, Outlier Monitoring, and Model Maintenance for Raman Spectra of Bioprocess Samples.
Zenodo. Version 0.1.1 [Computer software]. https://doi.org/10.5281/zenodo.23310422

Submitted Manuscripts

- **Ajayi, O. O.** (2026).
Label Noise and Scaffold Shift in Aqueous Solubility Prediction: Effects on the Accuracy and Uncertainty of Gaussian Processes, Random Forests, and Neural Networks.
Submitted to *ACS Omega*.
- **Ajayi, O. O.** (2026).
From Designed Mixtures to Fermentation Samples: Calibration Design, Outlier Monitoring and Model Maintenance for Raman PLS Models of Glucose and

Acetate.
Submitted to *Analyst*.

Manuscripts/Book Chapters in Preparation

- **Ajayi, O. O.**, Ajayi, V. O., & Aisedion, M.
  Artificial Intelligence and Multi-Omics in Oncology: Integrative Frameworks for Linking Micro(nano)plastic Exposure to Cancer Risk.
  In D. Jindal & M. M. Yallapu (Eds.), *Micro (nano) Plastics and Cancer Toxicology: Toxicological Mechanisms, Carcinogenesis and Therapeutic Implications*. Elsevier. (Chapter 10, in preparation.)
- **Ajayi, O. O.**, & Phillips, R. S.
  Recombinant Production and High-Throughput Crystallization Screening of the PLP-Dependent Aminotransferase AprL in Apramycin Biosynthesis
  (In preparation.)
- **Ajayi, O. O.**, et al.
  Pharmacophore-Guided Virtual Screening Identifies Candidate Inhibitors of AprL, a PLP-Dependent Aminotransferase in Apramycin Biosynthesis.
  (In preparation.)
- **Ajayi, O. O.**, & Ajayi, V. O.
  Comparison of Porous Graphitic Carbon (PGC), Hydrophilic Interaction Liquid Chromatography (HILIC), and Reversed Phase (RP) Columns.
  (In preparation.)